Technology Data Sheet (TDS)

TINMORRY ABS-Pro

Version 1.0

• Basic Introduction

TINMORRY ABS-Pro is a upgraded ABS with specially designed formula. It not only has excellent strength, toughness, temperature resistance, wear resistance, and durability, but also has significantly improved warping resistance, cracking resistance, and lighter odor than generic ABS. In addition, it can make the surface of the products smooth and delicate, and has a rich variety of colors to choose from, making it an ideal entry-level engineering filament.

Note: 1. It is more recommended to use an enclosed printer for printing, especially for large-sized and high-density models, so as to reduce warping and ensure sufficient layer strength. 2. It will produce a pungent odor during printing, please ensure good ventilation.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	1000 g ± 0.5%
Length	395 ± 5 m
Spool material	ABS ((Temperature resistance: 80 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	70 - 80 °C，6 - 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	≤ 65 ℃, ≤ 20% RH （Put in an air-tight container and sealed with desiccant, or put it in a filament drying box and set

	the temp to be about 60 °C ）
Storage temperature and humidity	≤ 30 °C, ≤ 20% RH（Sealed with desiccant）
Compatible support material	Itself or support filament for ABS
Compatible printer type	Enclosed-frame (recommended) Open-frame (not recommended)
Compatible nozzle material	Brass, stainless steel, hardened steel, etc
Compatible nozzle size	0.2, 0.4, 0.6, 0.8 mm
Compatible plate type	Smooth / Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	240 - 270 °C
Bed temperature	85 - 110 °C (depending on the printer and plate types)
Printing speed*	Max: 220 mm/s
Chamber temperature	< 65 °C

*When printed with Bambu X1C and P1S, and the nozzle temperature is 285 °C, the max printing speed can reach 300 mm/s.

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY ABS-Pro have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.05 g/cm^3
Saturated water absorption rate	25 °C , 55% RH	0.48%
Melt index	260 °C, 2.16 kg	8.7 ± 1.2 g/ 10 min
Glass transition temperature	DSC, 10 °C/min	N / A
Crystallization temperature	DSC, 10 °C/min	N / A
Melting temperature	DSC, 10 °C/min	204 °C
Vicar softening temperature	ISO 306, GB/T 1633	101 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	84 °C
Heat deflection temperature 2	ISO 75, 0.45 MPa	93 °C

• Mechanical Properties

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	35 ± 3 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	28 ± 2 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	2138 ± 183 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	2047 ± 168 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	8.2% ± 1.4%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	3.1% ± 0.8%

Bending strength (XY)	ISO 178, GB/T 9341	66 ± 2 MPa
Bending strength (Z)	ISO 178, GB/T 9341	45 ± 3 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	1964 ± 175 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	1875 ± 152 MPa
Impact strength (XY)	ISO 179, GB/T 1043	33.6 ± 2.2 kJ/m²
Impact strength (Z)	ISO 179, GB/T 1043	6.3 ± 1.4 kJ/m²

• Chemical and Other Properties

Subjects	**Data**
Component	ABS
Skin irritation	No
Resistance to water	Yes
Resistance to oil and grease	Not resistant to some kinds oil and grease for long-term contact
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Good
Resistance to strong acid	Not resistant
Resistance to weak alkali	Good
Resistance to strong alkali	Not resistant
Flammability	Yes
Odor during printing	Pungent
Odor during combustion	Strongly Pungent

• Specimen Info

Subjects	**Data**
Size	Shown at the following pictures
Nozzle temperature	270 °C
Bed temperature	90 °C
Printing speed	XY: 210 mm/s; Z: 100 mm/s
Infill density	100%
Infill pattern	Concentric

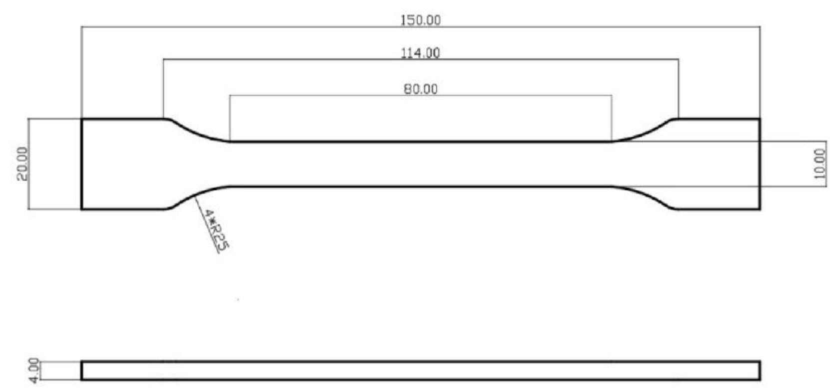
Statements：

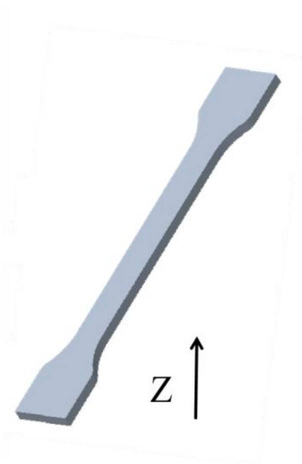
1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.

2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the

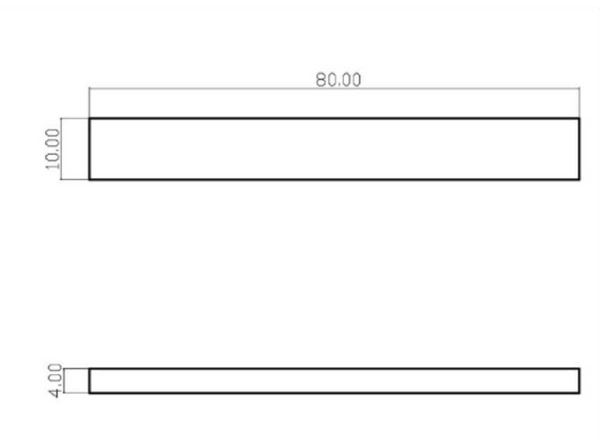
heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, comprehensive mechanical properties and heat resistance of TINMORRY ABS-Pro prints can be slightly increased, while some prints may undergo shrinkage or deformation, so please be aware of this.

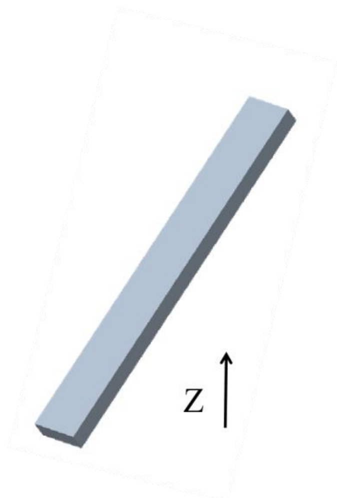
1. Tensile Test





2. Bending Test and Impact Test





• Disclaimer

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them,or the usage scenarios of them.